

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2}\right)\left(\frac{1}{N_1} + \frac{1}{N_2}\right)}}$$

Social Science Statistics

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2}\right)\left(\frac{1}{N_1} + \frac{1}{N_2}\right)}}$$

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Pearson Correlation Coefficient Calculator

The value of R is: 0.814.

Want to know how to report Pearson's r in your work? (Opens in a new tab so you don't lose your calculation.)

[How to report Pearson's \$r\$ \(APA\)](#)

Explanation of results

As you have probably already noticed, the output of this calculator is... verbose. Although most of the information provided below is self-explanatory, there are a few things worth noting. First, the five text boxes spread across the middle of the page represent the calculations that would be required if you were to calculate the R value in stages. Second, there is more than one way to calculate the R value, but these are all mathematically equivalent, so you shouldn't worry if you don't recognize the equation used here. Third, in the "Result Details & Calculations" box, you'll find what we've called a cross-check value, which is the R value calculated using an algorithm supplied by the [Meta Numerics](#) statistical library. This should be identical to the value that we've calculated.

Note: If you want to calculate a P value from your R score, [we have a calculator here](#) (before clicking, remember to note your r score and record any calculation details you require).

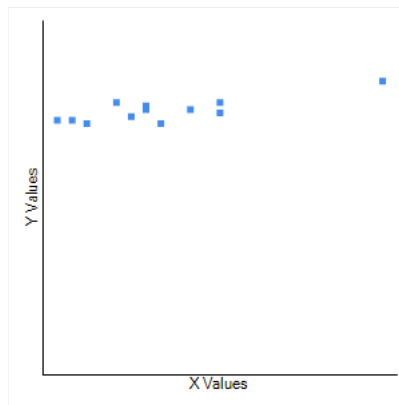


X Values

10.4
10.8
11.1
10.2
11.3
10.9
10.7
10.3
10.8
11.3
10.6
12.4

Y Values

7.1
7.6
7.5
7.2
7.4
7.1
7.3
7.2
7.5
7.7
7.7
8.3



X - M_x

-0.500
-0.100
0.200
-0.700
0.400
0.000
-0.200
-0.600
-0.100
0.400
-0.300
1.500

M_x : 10.900

Y - M_y

-0.367
0.133
0.033
-0.267
-0.067
-0.367
-0.167
-0.267
0.033
0.233
0.233
0.833

M_y : 7.467

(X - M_x)²

0.250
0.010
0.040
0.490
0.160
0.000
0.040
0.360
0.010
0.160
0.090
2.250

Sum: 3.860

(Y - M_y)²

0.134
0.018
0.001
0.071
0.004
0.134
0.028
0.071
0.001
0.054
0.054
0.694

Sum: 1.267

(X - M_x)(Y - M_y)

0.183
-0.013
0.007
0.187
-0.027
0.000
0.033
0.160
-0.003
0.093
-0.070
1.250

Sum: 1.800

Result Details & Calculation

Key

X Values

$$\Sigma = 130.8$$

$$\text{Mean} = 10.9$$

$$\Sigma(X - M_x)^2 = SS_x = 3.86$$

Y Values

$$\Sigma = 89.6$$

$$\text{Mean} = 7.467$$

$$\Sigma(Y - M_y)^2 = SS_y = 1.267$$

X and Y Combined

$$N = 12$$

$$\Sigma(X - M_x)(Y - M_y) = 1.8$$

R Calculation

$$r = \Sigma((X - M_x)(Y - M_y)) / \sqrt{((SS_x)(SS_y))}$$

$$r = 1.8 / \sqrt{((3.86)(1.267))} = 0.814$$

Meta Numerics (cross-check)

$$r = 0.814$$

X: X Values

Y: Y Values

M_x : Mean of X Values

M_y : Mean of Y Values

$X - M_x$ & $Y - M_y$: Deviation scores

$(X - M_x)^2$ & $(Y - M_y)^2$: Deviation Squared

$(X - M_x)(Y - M_y)$: Product of Deviation Scores

The value of R is 0.814.

This is a strong positive correlation, which means that high X variable scores go with high Y variable scores (and vice versa).

[Click here to calculate a p-value.](#)

The value of R^2 , the coefficient of determination, is 0.6626.

Calculate R

Reset